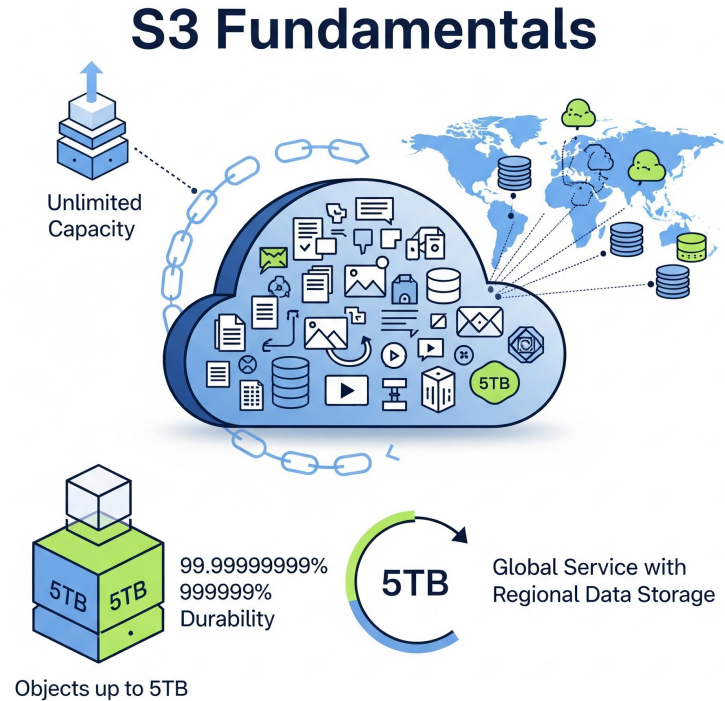


AWS Storage Services

S3 (Simple Storage Service)

S3 Fundamentals

- ❖ Object storage service
- ❖ Unlimited storage capacity
- ❖ Objects up to 5TB in size
- ❖ 99.999999999% durability
- ❖ Global service with regional data storage



S3 Key Concepts

- ❖ Buckets
- ❖ Objects



S3 Buckets

- ❖ Container for objects
- ❖ Globally unique name
- ❖ Region-specific
- ❖ Flat structure (no hierarchy)



Objects

- ❖ Files and metadata
- ❖ Key (name)
- ❖ Version ID
- ❖ Value (data)
- ❖ Metadata
- ❖ Subresources (ACLs, torrent)



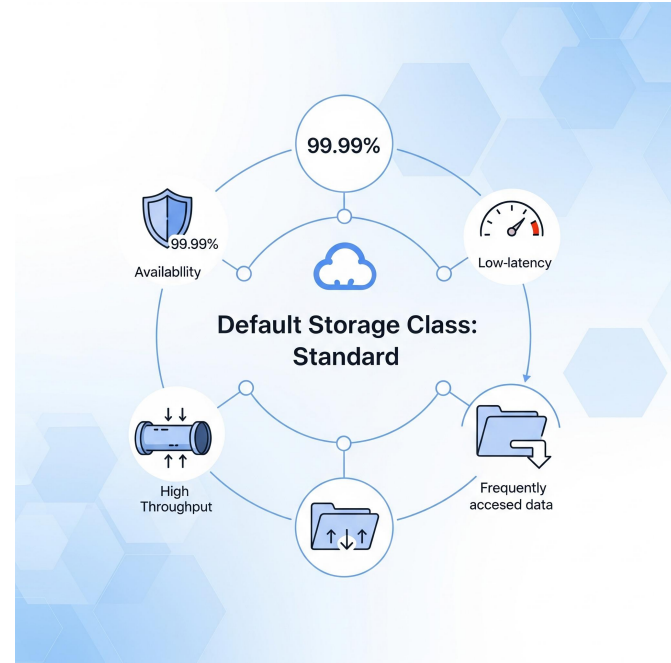
S3 Storage Classes

- ❖ S3 standard
- ❖ S3 Intelligent Tiering
- ❖ S3 Standard-IA(Infrequent Access)
- ❖ S3 one Zone-IA
- ❖ S3 Glacier
- ❖ S3 Glacier Deep Archive



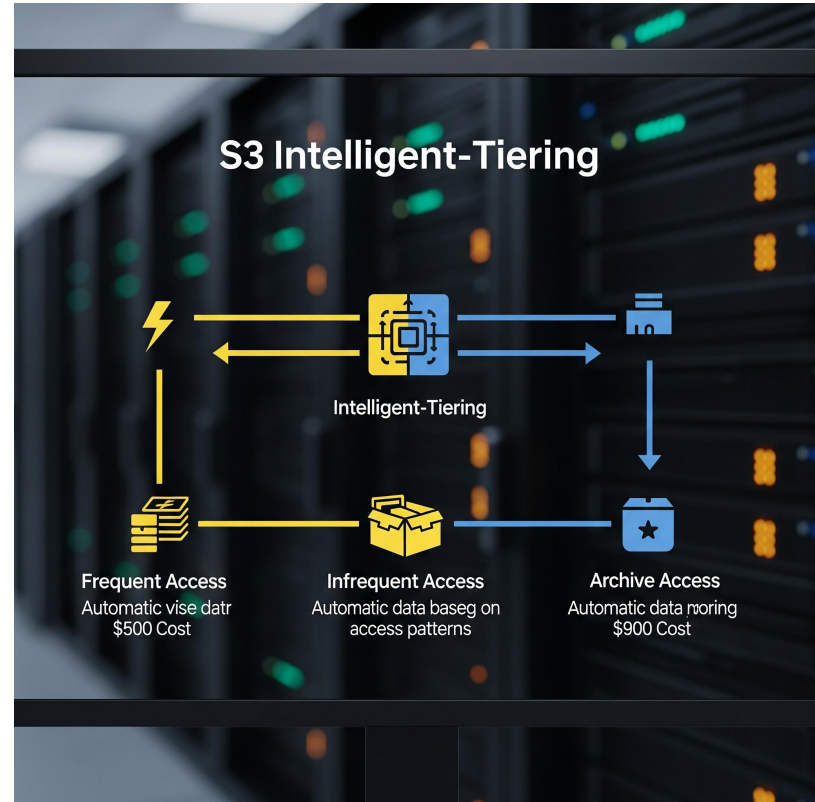
S3 Standard

- Default storage class
- 99.99% availability
- Low latency, high throughput
- Use cases: frequently accessed data



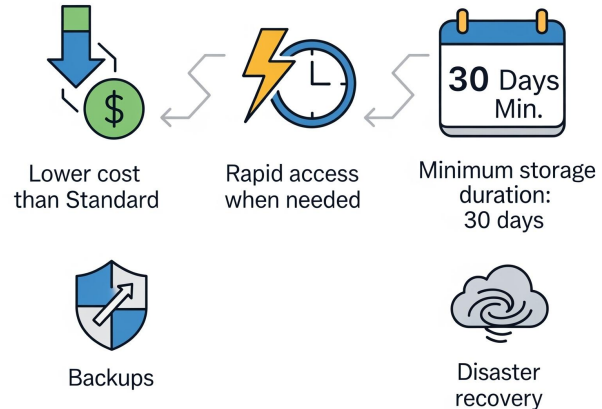
S3 Intelligent Tiering

- Automatic cost optimization
- Two access tiers
- Monthly monitoring fee
- Use cases: unpredictable access patterns



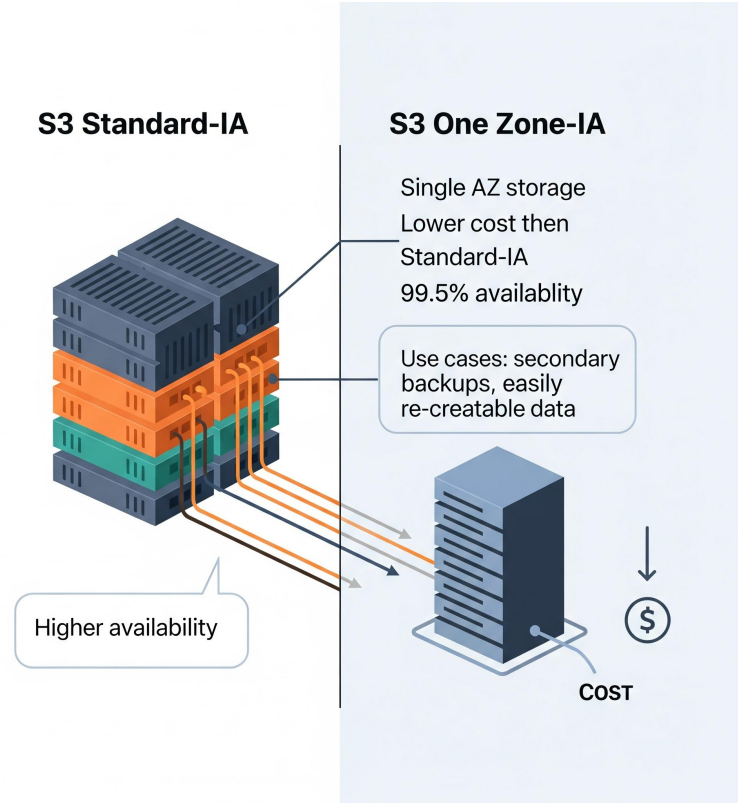
S3 Standard-IA (Infrequent Access)

- Lower cost than Standard
- Rapid access when needed
- Minimum storage duration: 30 days
- Use cases: backups, disaster recovery



S3 One Zone -IA

- Single AZ storage
- Lower cost than Standard-IA
- 99.5% availability
- Use cases: secondary backups



S3 Glacier

- Long-term archival storage
- Multiple retrieval options
- Minimum storage duration: 90 days
- Use cases: archival data



S3 Glacier Deep Archive

- Lowest-cost storage class
- 12-hour retrieval time
- Minimum storage duration: 180 days
- Use cases: long-term archival

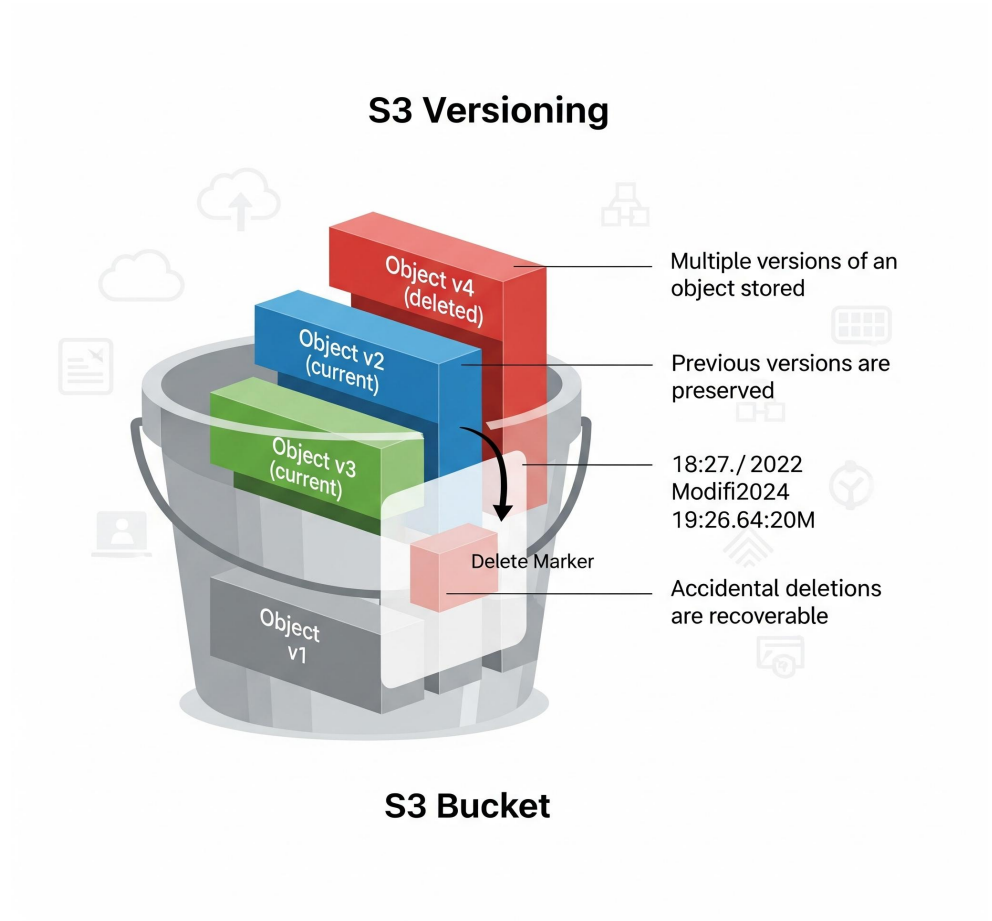
S3 Glacier Deep Archive



S3 features and capabilities

Versioning

- Multiple versions of objects
- Protection from deletions
- Roll back capabilities
- Version-specific permissions

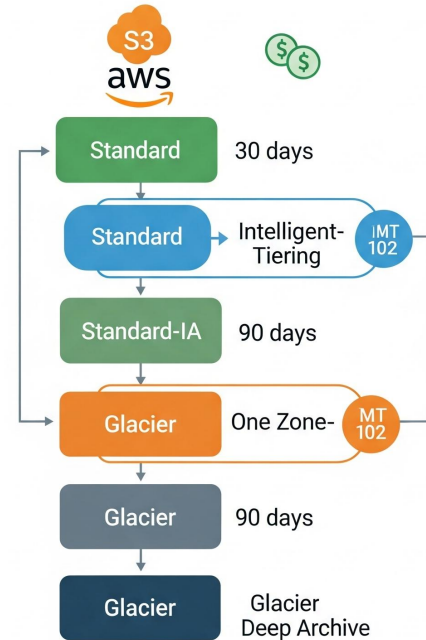


Lifecycle management

- Automated object transitions
- Expiration rules
- Version cleanup
- Cost optimization

S3 Lifecycle Management

Automated object transitions - Expiration rules - Version cleanup - Cost optimization



Expiration Rules

- Delete after 365 days

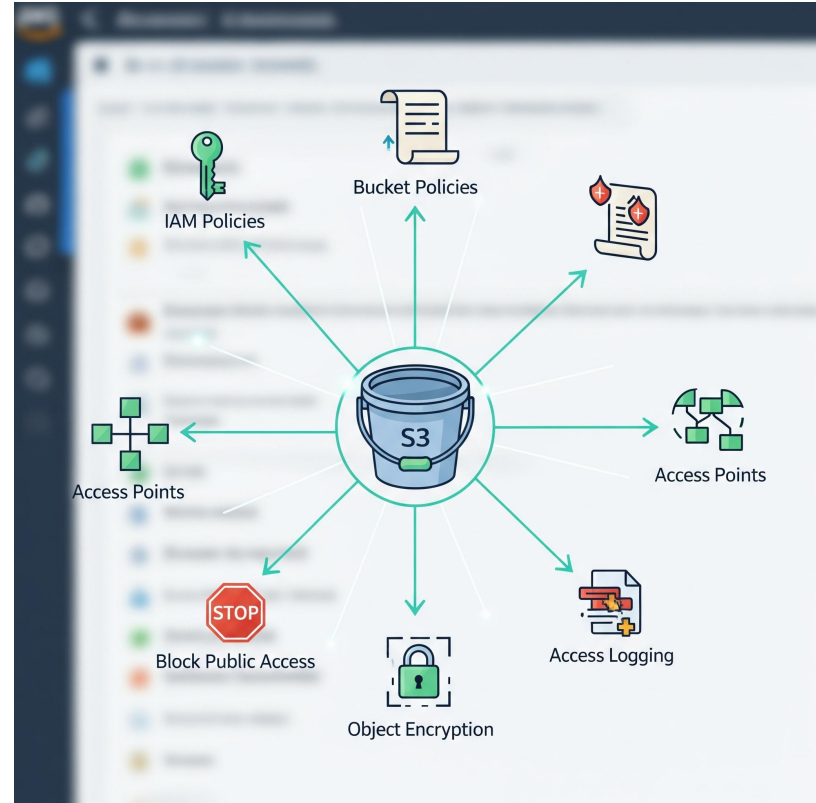


Version Cleanup



Security Features

- IAM policies
- Bucket policies
- Access Points
- Block Public Access
- Object encryption
- Access logging



Data Transfer

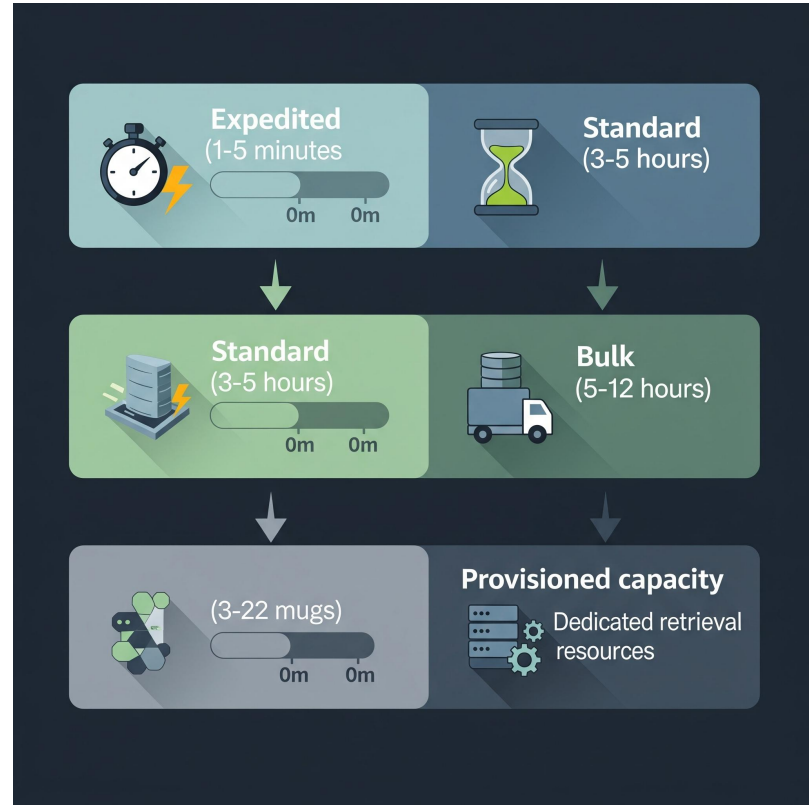
- Transfer Acceleration
- Multipart upload
- Cross-Region Replication
- Same-Region Replication
- Event notifications



S3 Glacier Deep Dive

Retrieval Options

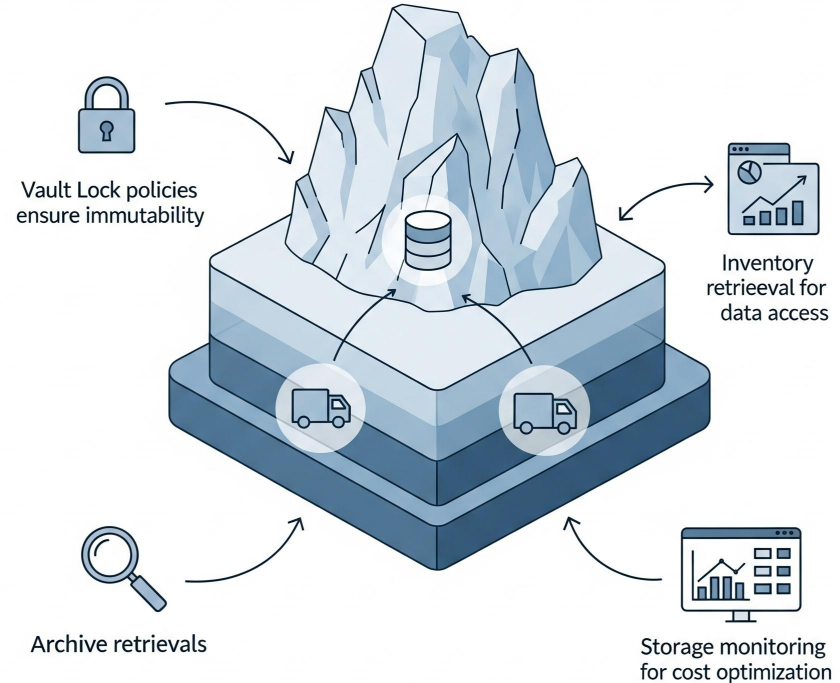
- Expedited (1-5 minutes)
- Standard (3-5 hours)
- Bulk (5-12 hours)
- Provisioned capacity



Data Management

- Vault Lock policies
- Inventory retrieval
- Archive retrievals
- Storage monitoring

Data Management with S3 Glacier



EBS (Elastic Block Store)

EFS

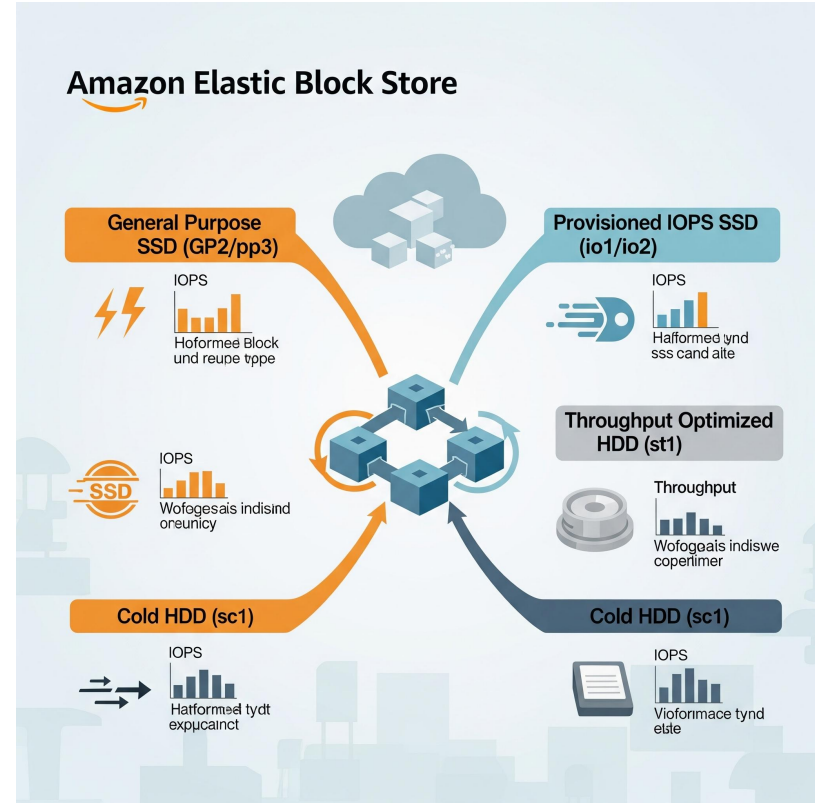
EBS Key Features

- Point-in-time snapshots
- Encryption at rest
- Dynamic volume modification
- Multi-attach capability



EBS Volume Types

- General Purpose SSD (gp2/gp3)
- Provisioned IOPS SSD (io1/io2)
- Throughput Optimized HDD (st1)
- Cold HDD (sc1)



Performance

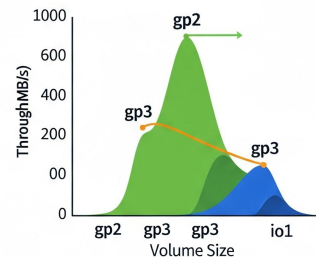
- IOPS allocation
- Throughput limits
- Volume size considerations
- Burst credits (gp2)

EBS Performance

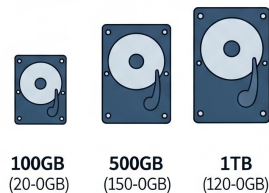
IOPS Allocation



Throughput Limits



Volume Size Considerations



Burst Credits (gp2)



Burst credits are earned & earned
bused and consumed

Advance EBS Features

- ❖ Snapshots
- ❖ Encryption
- ❖ Operations



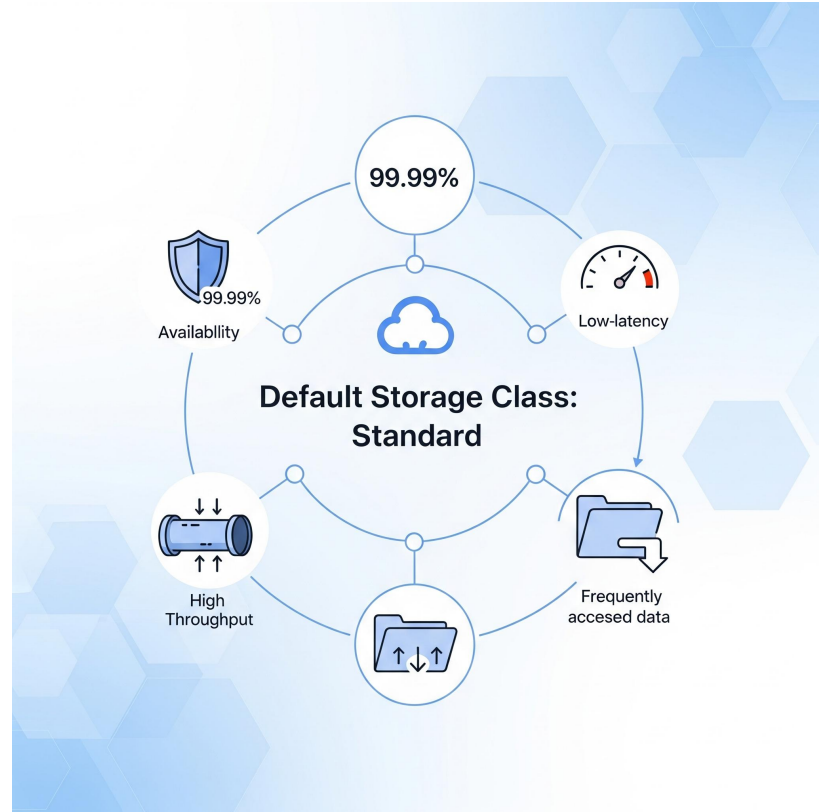
Snapshots

- Incremental backups
- Cross-region copying
- Sharing capabilities
- Automated lifecycle



Encryption

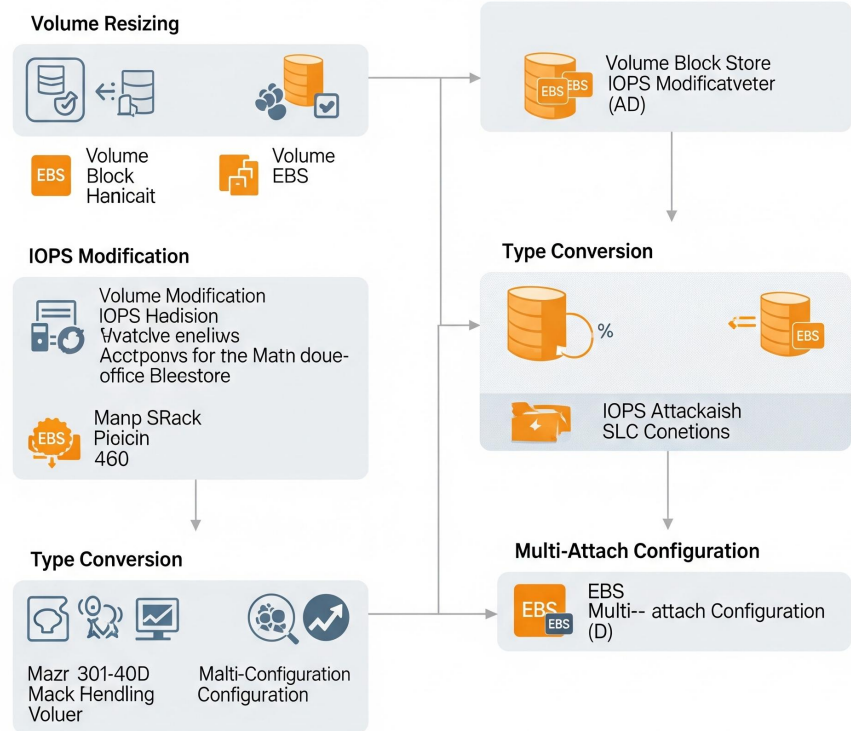
- AWS KMS integration
- Default encryption
- Key rotation
- Cross-region replication



Operations

- Volume resizing
- IOPS modification
- Type conversion
- Multi-attach configuration

Elastic Block Store



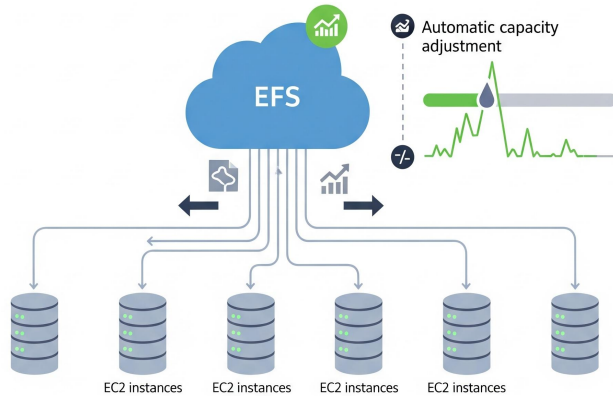
EBS Encryption



- AWS KMS integration for secure encryption
- Default encryption for new EBS volumes
- Automated key rotation enhances security
- Cross-region replication of encrypted snapshots

EFS

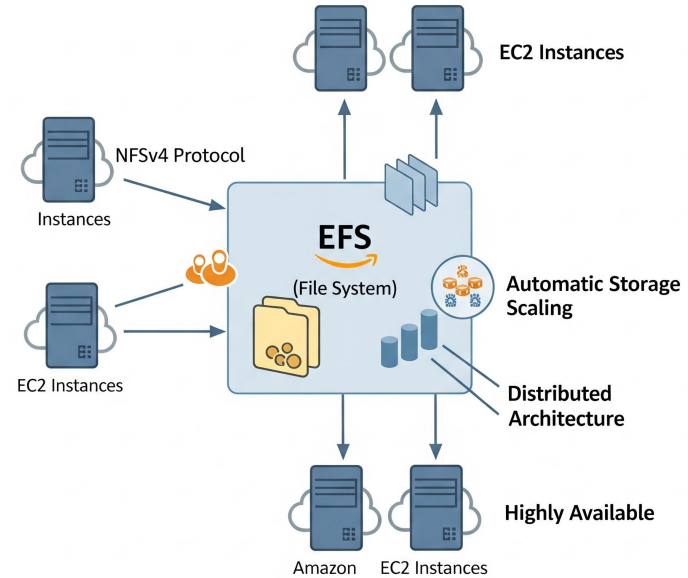
EFS Definition



- ❖ Scalable network file system for AWS
- ❖ Shared access across EC2 instances
- ❖ Supports NFSv4 protocol
- ❖ Automatically grows and shrinks capacity
- ❖ File-based (POSIX-compliant)
- ❖ Scales automatically with data size
- ❖ Supports multi-AZ availability

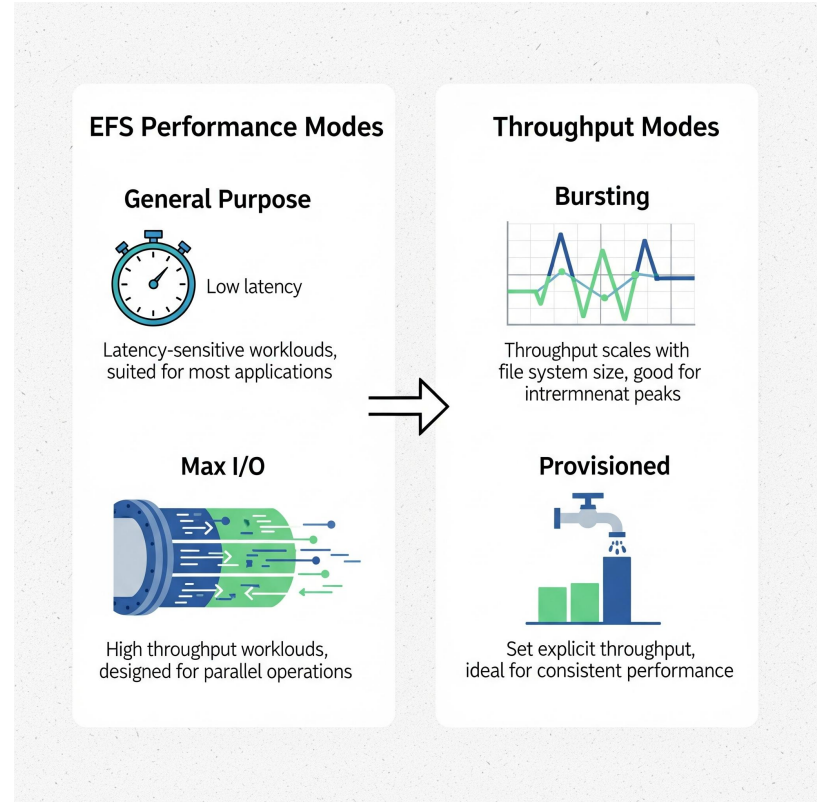
EFS Architecture

- ❖ Shared access for multiple EC2 instances
- ❖ Supports NFSv4 protocol for compatibility
- ❖ Automatically scales storage capacity as needed
- ❖ Distributed and highly available architecture



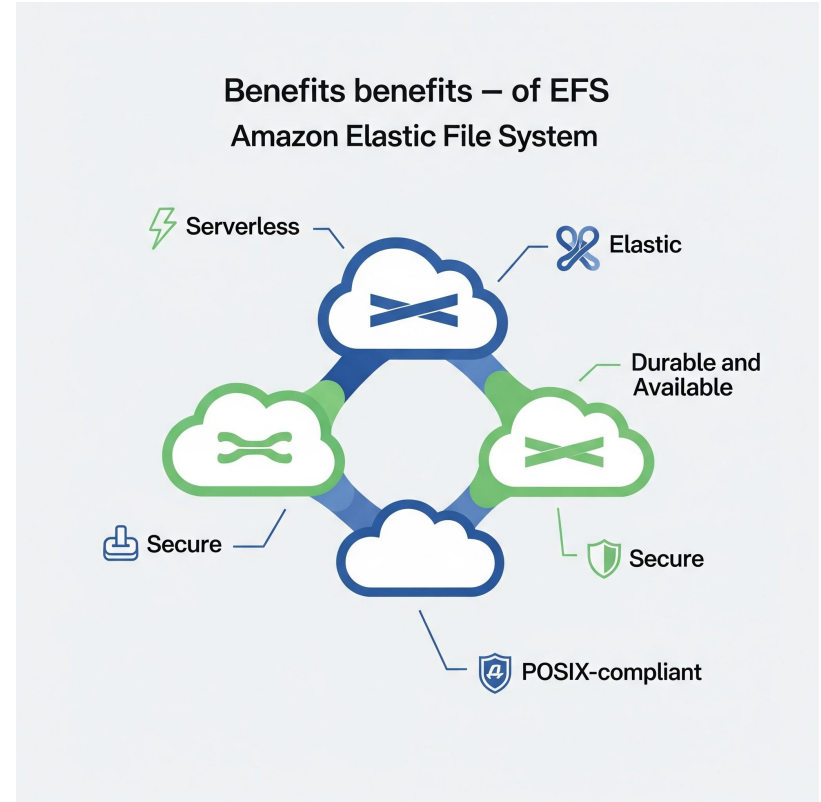
EFS Performance and Throughput Modes

- ❖ Performance Modes:
 - General Purpose: Best for latency-sensitive use cases
 - Max I/O: Optimized for high throughput at the cost of latency
- ❖ Throughput Modes:
 - Bursting: Based on file system size
 - Provisioned: Explicitly set throughput regardless of storage size



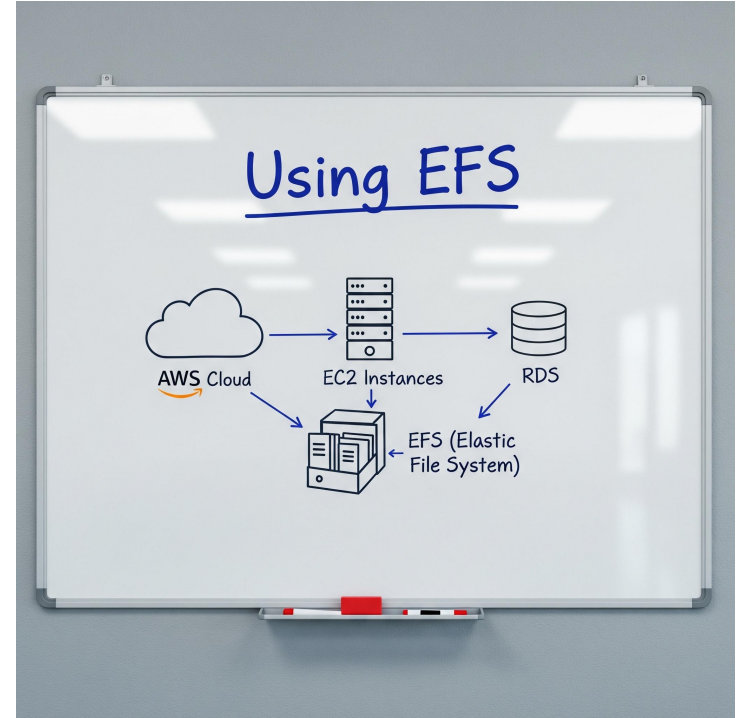
Benefits of EFS

- ❖ Serverless: No provisioning, fully managed
- ❖ Elastic: Scales automatically
- ❖ Durable and Available: Data stored across multiple AZs
- ❖ Secure: Supports IAM, encryption at rest and in transit
- ❖ POSIX-compliant: Familiar file access semantics



Using EFS

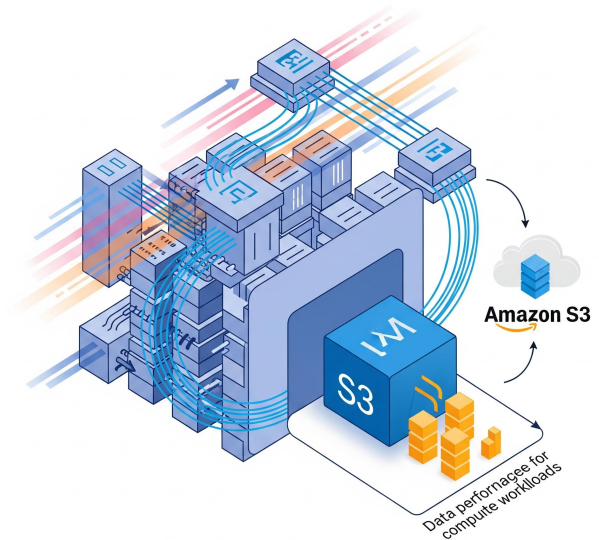
- ❖ **Create EFS File System:** Choose performance and throughput modes, and enable encryption if needed.
- ❖ **Configure Mount Targets:** Add mount targets in each AZ/subnet where EC2 instances reside.
- ❖ **Mount EFS on EC2:** Install `amazon-efs-utils` or use an NFS client, then mount the file system.
- ❖ **Manage Access:** Use Security Groups, IAM policies, and POSIX permissions.
- ❖ **Monitor Usage:** Use CloudWatch metrics for throughput, IOPS, and burst credit.



FSx

FSx for Lustre

- ❖ High-performance file system for compute workloads
- ❖ Optimized for fast processing of data
- ❖ Integrates with Amazon S3 for data access
- ❖ Supports various deployment options



FSx Lustre Use Cases

- ❖ Optimized for High Performance Computing workloads
- ❖ Ideal for machine learning and analytics
- ❖ Seamlessly integrates with Amazon S3 data lakes
- ❖ Supports various deployment and storage options

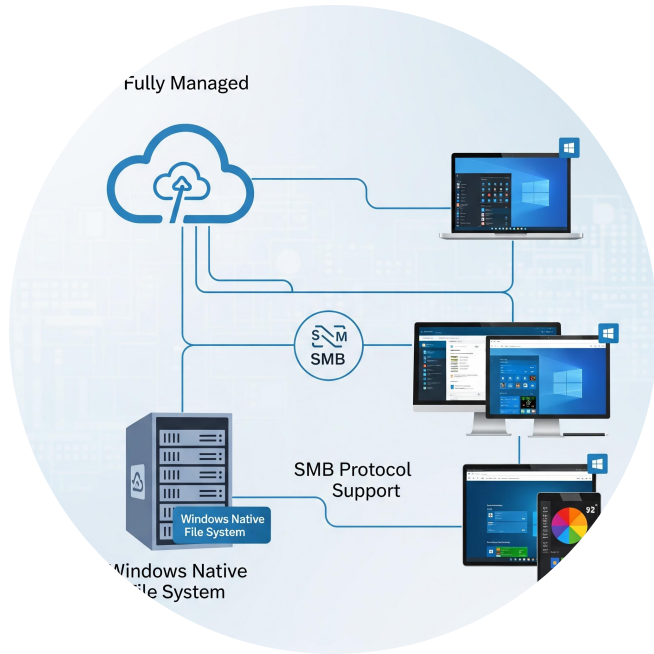


FSx Lustre Benefits

- ❖ High-performance for demanding compute workloads
- ❖ Optimized for machine learning and analytics
- ❖ Seamlessly integrates with Amazon S3 data
- ❖ Supports various deployment and storage options



FSx Windows Features

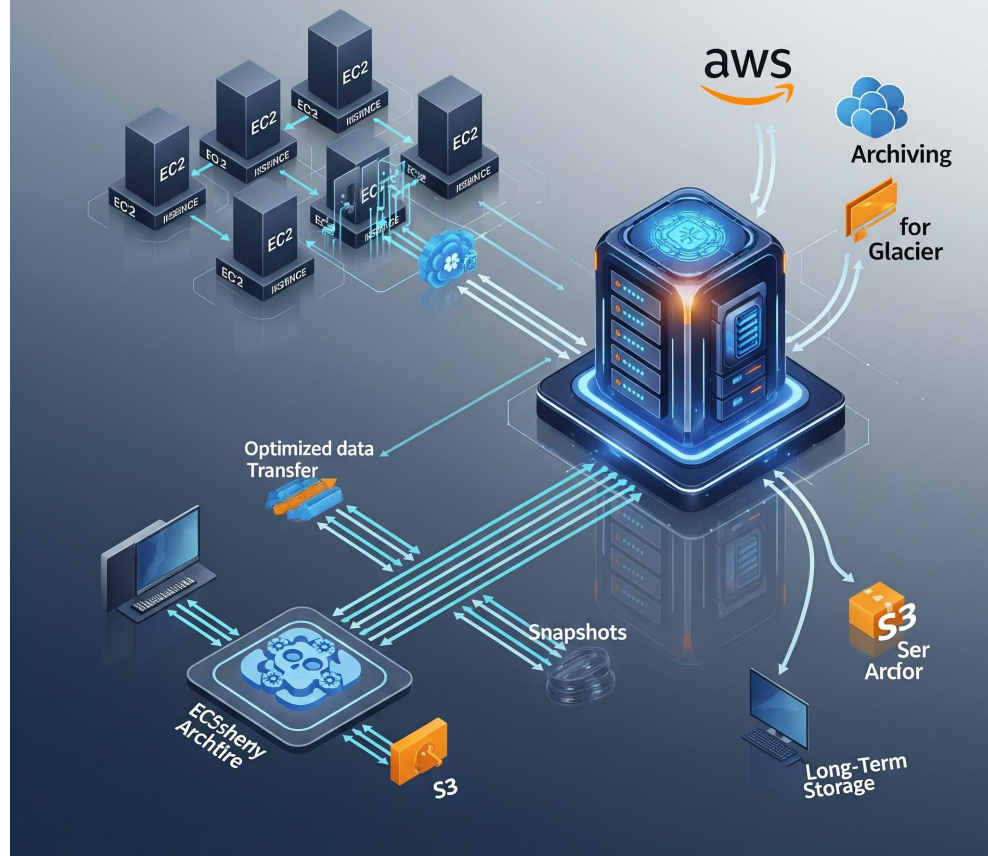


- ❖ Fully managed Windows native file system
- ❖ Supports SMB protocol for Windows applications
- ❖ Integrates with existing Active Directory
- ❖ High availability across multiple Availability Zones

FSx for OpenZFS

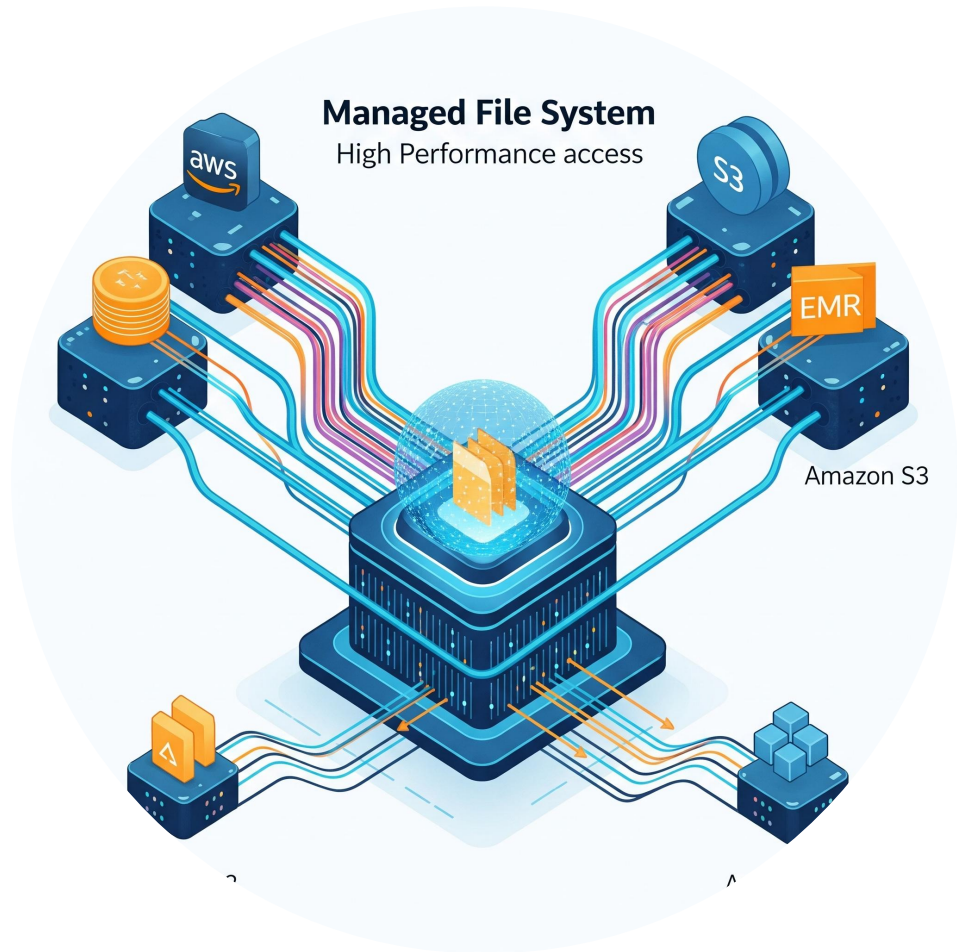
- ❖ Fully managed native OpenZFS file system
- ❖ Supports ZFS features like snapshots and replication
- ❖ High performance for demanding workloads
- ❖ Integrates with AWS services

Fully Managed Native OpenZFS File System



FSx for NetApp ONTAP

- ❖ Fully managed NetApp ONTAP file system
- ❖ Supports all ONTAP features and APIs
- ❖ High performance and multi-protocol access
- ❖ Integrates with other AWS services



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- | | | | |
|----|--|----|--|
| 01 | Fully managed NetApp ONTAP file system | 03 | Supports all ONTAP features and APIs |
| 02 | Integrates with other AWS services | 04 | High performance and multi-protocol access |

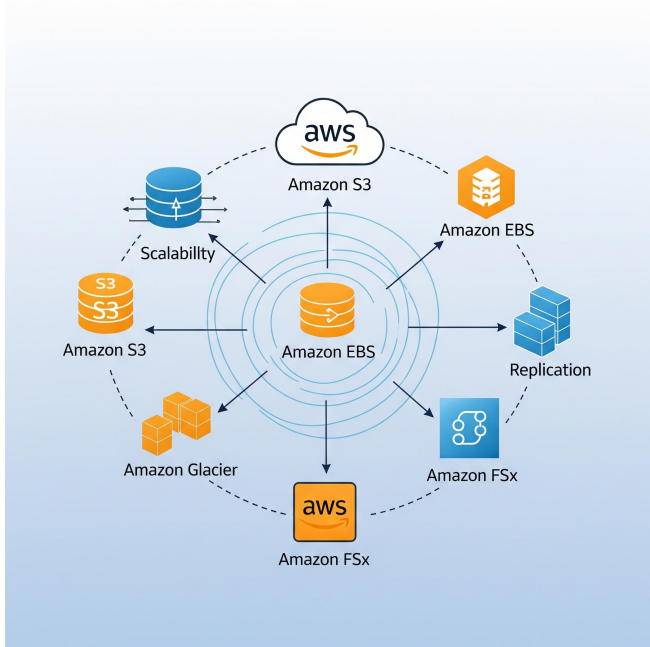
Benefits of FSx for NetApp ONTAP

- ❖ Fully managed NetApp ONTAP file system
- ❖ Supports all ONTAP features and APIs
- ❖ High performance and multi-protocol access
- ❖ Integrates with other AWS services

Benefits of FSx for NetApp ONTAP

- ❖ Fully managed NetApp ONTAP file system
- ❖ Supports all ONTAP features and APIs
- ❖ High performance and multi-protocol access
- ❖ Integrates with other AWS services

Key Takeaways



- Understand core features of AWS storage
- Match services to specific use cases
- Optimize cost with various storage classes
- Leverage security and data management features

Knowledge Check-1

Which of the following is NOT a characteristic of Amazon S3 (Simple Storage Service)?

- a) Unlimited storage capacity
- b) Objects up to 5TB in size
- c) 99.999999999% durability
- d) Primarily used for relational databases**

Knowledge Check-2

Which of the following is a key characteristic of Amazon S3 Intelligent Tiering?

- a) It is the lowest-cost storage class for long-term archival.
- b) It requires manual intervention to move data between access tiers.
- c) It automatically optimizes storage costs by moving data between two access tiers based on access patterns.
- d) It has a minimum storage duration of 180 days.

Summary

01 Summarizes the diverse AWS storage services

03 Highlights key features and benefits of each

02 Essential for understanding storage solutions

04 Provides an overview of various use cases

